

AMENDMENT NO. 1 JULY 2024

TO

IS 16202 : 2014 AGRO TEXTILES — WOVEN GROUND COVERS FOR HORTICULTURE APPLICATION — SPECIFICATION

(Foreword, para 5) — Substitute the following for the existing:

‘The composition of the Committee responsible for the formulation of this standard is given in Annex E.’

(Page 1, clause 4.1, line 3) — Substitute the following for the existing:

‘The linear density of polypropylene tape when measured as per Annex D shall be either 550 denier for warp and 1 000 denier for weft or 450 denier for warp and 1 000 denier for weft.’

(Page 1, clause 4.1, para 1) — Insert the following sentence at the end:

‘The denier of polypropylene tape used in the manufacture of woven ground cover shall be subjected to the following tolerances:

- a) ± 10 percent on individual value; and
- b) ± 5 percent on average.’

(Page 1, clause 4.2, para 1) — Substitute the following for the existing:

‘The ground cover shall be woven with plain weave and a sett of either 80 ends/dm and 40 picks/dm or a sett of 96 ends/dm and 40 picks/dm. The tolerance on ends/dm and picks/dm shall be ± 2 .

NOTE — The combination of sett and denier selected for woven ground cover shall meet the requirement of GSM (g/m^2) specified in Table 1.’

(Page 1, clause 5.2) — Delete the entry for tolerance on mass (g/m^2).

[Page 2, Table 1, Sl No. (ii), (iii), (iv), and (vi)] — Substitute the following for the existing:

Sl No.	Characteristic(s)	Requirement(s)	Method of Test, Ref to
(1)	(2)	(3)	(4)
ii)	Tensile strength (grab method, 200 mm* 100 mm, 25 mm gauge width), kgf, <i>Min</i>		IS 1969 (Part 2)
	a) Warp way	65	
	b) Weft way	35	
iii)	Trapezoid tear strength, kgf, <i>Min</i>		IS 14293
	a) Warp way	30	
	b) Weft way	18	
iv)	Air permeability, mm/s, <i>Min</i>	150	IS 11056
vi)	UV accelerated exposure testing, percent strength retained, <i>Min</i>		Annex C
	a) Warp way	70	
	b) Weft way	70	

Price Group 2

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(Page 5, clause C-3.4) — Substitute the following for the existing formula:

$$\text{'Percent retention of original breaking strength} = \frac{b}{a} \times 100'.$$

(Page 5, Annex C) — Insert the following new annex after Annex C:

'ANNEX D

(Clause 4.1)

METHOD OF TEST FOR MEASUREMENT OF LINEAR DENSITY (DENIER) OF TAPE

D-1 PRINCIPLE

The linear density is determined from the mass of a specified length of tape and expressed in denier or tex. The specimen is first laid straight, free from any twist and wrinkles and then the length is measured. The specimen is then weighed, and the mass is used in calculating the linear density (denier) of the specimen.

D-2 APPARATUS

D-2.1 Digital Balance with Pan

A digital balance with a sample holding pan, capable of weighing test specimens to an accuracy of 0.001 g.

D-2.2 Length Measuring Scale

Metallic ruler or a suitable length measuring scale, graduated in millimeters, and capable of measuring dimensions up to 10 m with an accuracy of 1 mm.

D-3 PREPARATION OF TEST SPECIMENS

Select a minimum of five sample tape packages, discard the initial few metres of tape from the sample package. From each tape package, draw enough length of tape specimen and keep it on a flat horizontal table with markings at both ends, with a specified dimension between the two marks, preferably 9 m for denier and 10 m for tex measurement, marking may be adjusted according to the table length. Render the tape yarn flat on a table, free from any twists or wrinkles. Measure and cut the tape specimen between the two marks. Measure a minimum of five tape specimens from each package.

D-4 TEST PROCEDURE

For determination of mass of the test specimen (M), preferably take 9 m length of tape specimen (L) for denier or 10 m length of tape specimen (L) for tex measurement. Measure the specimen length to an accuracy of 1 mm and correspondingly weigh each test specimen correct to 0.001 g.

D-5 CALCULATIONS

Calculate the linear density for each test specimen by one of the following equations:

$$\text{Denier} = \frac{100 \times 90 \times M}{L}$$

$$\text{Tex} = \frac{100 \times 10 \times M}{L}$$

where

M = mass, in g, of the test specimen; and

L = length, in m, of test specimen.

D-6 TEST RESULTS

Calculate denier/tex for a minimum five samples from each package and calculate the average for each individual package. A tolerance of ± 10 percent on the individual value (25 values) and ± 5 on average (5 values) for each package shall be permissible.'

(Page 6, Annex D, Heading) — Substitute 'Annex E' for 'Annex D'.